WORKED SOLUTION

STEP III 2005, Question 4

Pure • Non-linear Recurrence Relations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q4

The key idea

The recurrence is not linear, but it is really about *ratios* of consecutive terms. If we write $u_{n+1} = \rho\, u_n$, the recurrence says the next ratio is $k - \rho$. So the ratios flip back and forth between two values whose sum is k, and everything else follows from that. The three special cases then become conditions on those two ratios.

Solution

The induction

Write $r = \frac{b}{a}$. Dividing the recurrence by u_{n+1} (when it is non-zero) gives

$$\frac{u_{n+2}}{u_{n+1}} = k - \frac{u_{n+1}}{u_n},$$

so each ratio of consecutive terms is k minus the previous ratio. The first ratio is $u_2/u_1 = r$, so the next should be

$$c = k - \frac{b}{a} = \frac{ka - b}{a}.$$

Note that $c \neq 0$, because $b \neq ka$. Also $r \neq 0$, because $b \neq 0$. Together these keep every term non-zero, so we never divide by zero.

Let $P(n)$ be the statement that $u_{2n-1} \neq 0$ and $u_{2n} = r\, u_{2n-1}$. $P(1)$ holds, since $u_1 = a \neq 0$ and $u_2 = b = ra$.

Suppose $P(n)$ holds for some $n \geq 1$. Then $u_{2n} = ru_{2n-1} \neq 0$, and the recurrence gives

$$u_{2n+1} = \frac{u_{2n}}{u_{2n-1}}(ku_{2n-1} - u_{2n}) = r\,(k - r)\,u_{2n-1} = (k - r)\,u_{2n} = c\,u_{2n},$$

which is the second required relation. Hence $u_{2n+1} \neq 0$, and

$$u_{2n+2} = \frac{u_{2n+1}}{u_{2n}}(ku_{2n} - u_{2n+1}) = c\,(k - c)\,u_{2n} = (k - c)\,u_{2n+1} = r\,u_{2n+1},$$

because $k - c = r$. That is $P(n+1)$. By induction, for every $n \geq 1$,

$$u_{2n} = \frac{b}{a}\, u_{2n-1}, \qquad u_{2n+1} = c\, u_{2n}, \qquad c = k - \frac{b}{a}.$$

Ask yourself Why put "$u_{2n-1} \neq 0$" into the induction statement?

The recurrence divides by u_n, so the sequence only makes sense if no term is zero. Carrying non-zeroness through the induction proves the terms are defined at the same time as it proves the formulae. This is where both conditions $b \neq 0$ and $b \neq ka$ are used.

Closed forms

Going two steps multiplies by c and then by $\frac{b}{a}$, so $u_{2n+1} = \frac{bc}{a}\, u_{2n-1}$. The odd terms form a geometric progression with first term a and ratio $\frac{bc}{a}$, and each even term is $\frac{b}{a}$ times the odd term before it:

$$u_{2n-1} = a\Big(\frac{bc}{a}\Big)^{n-1} = \frac{b^{n-1}c^{n-1}}{a^{n-2}}, \qquad u_{2n} = b\Big(\frac{bc}{a}\Big)^{n-1} = \frac{b^n c^{n-1}}{a^{n-1}}.$$

Part (i)

The ratios of consecutive terms alternate $\frac{b}{a}, c, \frac{b}{a}, c, \ldots$, so the sequence is geometric exactly when $c = \frac{b}{a}$, that is $k - \frac{b}{a} = \frac{b}{a}$:

$$ka = 2b.$$

The common ratio is then $\frac{b}{a}$. This is consistent with $b \neq ka$, because $ka = 2b \neq b$ when $b \neq 0$.

Part (ii)

From the closed forms, $u_{n+2} = \frac{bc}{a}\, u_n$ for every n. So $u_{n+2} = u_n$ for all n exactly when $\frac{bc}{a} = 1$, that is $b(ka - b) = a^2$:

$$kab = a^2 + b^2.$$

The sequence is then $a, b, a, b, \ldots$. If we want the period to be exactly 2 (not a constant sequence) we also need $a \neq b$. If $a = b$ the condition becomes $k = 2$, and every term equals a.

Part (iii)

Similarly $u_{n+4} = (\frac{bc}{a})^2 u_n$, so $u_{n+4} = u_n$ for all n exactly when $(\frac{bc}{a})^2 = 1$, that is $\frac{bc}{a} = \pm 1$. The value $+1$ is part (ii), where the period is 2 (or 1). For a period of exactly 4 we need $\frac{bc}{a} = -1$, that is $b(ka - b) = -a^2$:

$$kab = b^2 - a^2.$$

The sequence is then $a, b, -a, -b, a, b, \ldots$. Since $a \neq 0$ we have $-a \neq a$, so it really does have period 4 and not less.

In both (ii) and (iii) the condition $b \neq ka$ holds automatically. Putting $b = ka$ into either condition gives $a^2 = 0$, which is impossible.

Checks

Try $a = 1$, $b = 2$. With $k = 4$ (part (i)) the recurrence gives $1, 2, 4, 8, 16, \ldots$. With $k = \frac{5}{2}$ (part (ii)) it gives $1, 2, 1, 2, \ldots$. With $k = \frac{3}{2}$ (part (iii)) it gives $1, 2, -1, -2, 1, 2, \ldots$. We also ran the recurrence in exact rational arithmetic for about 2000 random choices of a, b and k. Every one matched the closed forms, and the sequence was geometric, 2-periodic or 4-periodic exactly when the conditions above held.

Common slips

- Forgetting to say why no term is ever zero, when the recurrence divides by u_n.
- Giving $\frac{bc}{a} = \pm 1$ as the answer to (iii). The $+1$ case has period 2.
- In (ii), not noticing that $a = b$, $k = 2$ gives a constant sequence.

Beyond STEP

The substitution $\rho_n = u_{n+1}/u_n$ turns this non-linear recurrence into the first-order map $\rho \mapsto k - \rho$. That map is an involution: applying it twice gets you back where you started, which is why the period is 2 in the ratios. Changing variables to make a non-linear recurrence or differential equation simpler is a standard first-year technique, and the study of periodic orbits of maps begins discrete dynamical systems. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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